

Simok Lee

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Education

- Mar. 2021 – Present: M.S.-Ph.D. Integrated, Electrical Engineering
Korea Advanced Institute of Science and Technology (KAIST)
Daejeon, Republic of Korea
Advisor: Prof. Jae-Woong Jeong
 - Mar. 2016 – Feb. 2021: B.S., Chemical and Biomolecular Engineering
Korea Advanced Institute of Science and Technology (KAIST)
Daejeon, Republic of Korea
Minor in Electrical Engineering
Cum Laude
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Experience

- Jul. 2024 – Dec. 2024: Visiting Researcher, Department of Chemical and Biomolecular Engineering
North Carolina State University, USA
Advisor: Prof. Michael Dickey
 - Mar. 2019 – Aug. 2019: Exchange Program
Technical University of Darmstadt, Germany
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Awards and Honors

1. **Best PhD Research Achievement Award** by KAIST School of Electrical Engineering (Mar. 2026)
 2. **Next-Generation Engineer Award (First Place)** by Institute for Promotion of Engineering and Science of Korea (IPESK) (Nov. 2025)
 3. Outstanding Teaching Assistant Award by KAIST Electrical Engineering (Nov. 2025)
 4. All Scientific Poster Vendor Prize by 2024 Carolina Science Symposium (Nov. 2024)
 5. Outstanding Paper Award by 2024 The Korean Sensors Society Spring Conference (Apr. 2024)
 6. Young-Han Kim Global Leader Scholarship by KAIST (Jul. 2023)
(Received 4000 USD)
 7. **Best Poster Award by 2023 MRS Spring Meeting** (Apr. 2023)
(Awarded to 18 out of 1382 posters, 1.3%)
 8. Best Paper Award by 2022 The Korean Sensors Society Autumn Conference (Aug. 2022)
 9. Excellent Award for 2020 Summer/Fall Undergraduate Research Program (URP) by KAIST (Feb. 2021)
 10. Cum Laude by KAIST (Feb. 2021)
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Publications ([Google Scholar](#))

Journal Papers ([†] These authors contributed equally)

Papers Under Review

1. **Simok Lee**†, Min Seok Kim†, Se-Yeon Heo, Woo Jae Kim, Yejin Ahn, Juhyeong Lee, Do A Kwon, Subin Oh, Sung-Eui Yoon, Michael D. Dickey, Young Min Song*, Jae-Woong Jeong*, “Liquid-Metal Physical Unclonable Functions Resistant to Both Physical and Digital Cloning”, *Nature Communications* (Under Review)
2. Yejin Ahn†, **Simok Lee**†, Choong Yeon Kim, and Jae-Woong Jeong*, “Skin-Like Blind-Spot-Free Tactile Sensing via Phase-Programmable Multimode Liquid Metal Pressure Sensors”, *npj Flexible Electronics* (Under Review)
3. Byungkook Oh, Soomin Kim, Young-Soo Lim, Chaewon Gong, Seok Hyun Hong, Congqi Yang, Dong Jun Kim, Minseo Kang, **Simok Lee**, Kum Seok Nam, Jae-Woong Jeong, Taek-Soo Kim, Sumin Kang, Seongjun Park, Seungbum Hong, Sung-Min Park, Steve Park*, “Vacuum-Processable Hydrogel for Zero-Gauge Factor Stretchable Gold Electronics”, *Nature Materials* (Under Review)

Published Papers

1. Subin Oh, **Simok Lee**, Sung Woo Kim, Yejin Ahn, Dongho Min, Semin Kim, and Jae-Woong Jeong, “Liquid Metals for Reconfigurable Bioelectronics”, *Advanced Materials*, e21174, Mar. 2026. ([Link](#))
2. Mohammadreza Zare, Man Hou Vong, Mohamed Irfan, **Simok Lee**, Cordelia McKelvy, Febby Krisnadi, Michael A. Daniele, Qingshan Wei, Michael D. Dickey*, “Capture and Release of Molecules with Liquid Metals,” *ACS Applied Materials & Interfaces*, 18, 3038-3051, Jan. 2026. ([Link](#))
3. **Simok Lee**, Jinwoo Ma, Yasunori Hioki, Kosuke Nakano, Koji Yoshida, Mark Waugh, Mohammadreza Zare, Jae-Woong Jeong, Michael D. Dickey*, “Practical Considerations for Gallium-based Liquid Metal for Stretchable Electronics: Metal Contacts and Strain Cycling,” *ACS Applied Electronic Materials*, 7, 10610-10619, Oct. 2025. ([Link](#))
(Featured on the Front Cover)
4. **Simok Lee**†, Gun-Hee Lee†, Choong Yeon Kim, Inho Kang, Steve Park*, and Jae-Woong Jeong*, “Phase-change metal ink with pH-controlled chemical sintering for versatile and scalable fabrication of variable stiffness electronics,” *Science Advances* 11, eadv4921, May 2025 ([Link](#))
5. **Simok Lee**, Syed Ahmed Jaseem, Nurit Atar, Meixiang Wang, Jeong Yong Kim, Mohammadreza Zare, Sooyoung Kim, Michael D. Bartlett, Jae-Woong Jeong, Michael D. Dickey*, “Connecting the Dots: Sintering of Liquid Metal Particles for Soft and Stretchable Conductors,” *Chemical Reviews* 125, 3551, Mar. 2025. ([Link](#))
(Featured on the Front Cover)
6. Subin Oh†, **Simok Lee**†, Sung-Woo Kim, Choong Yeon Kim, Eun Young Jeong, Juhyun Lee, Do A Kwon, and Jae-Woong Jeong*, “Softening implantable bioelectronics: Material designs, applications, and future directions,” *Biosensors and Bioelectronics*, 116328, Aug. 2024. ([Link](#))
7. Do A Kwon†, **Simok Lee**†, Choong Yeon Kim, Inho Kang, Steve Park*, and Jae-Woong Jeong*, “Body-temperature softening electronic ink for additive manufacturing of transformative bioelectronics via direct writing,” *Science Advances* 10, eadn1186, Feb. 2024. ([Link](#))
(Review paper published in **MATTER** (Cell Press) highlighting this work ([Link](#)))

8. Karen-Christian Agno†, Keungmo Yang†, Sang-Hyuk Byun, Subin Oh, **Simok Lee**, Heesoo Kim, Kyurae Kim, Sungwoo Cho, Won-Il Jeong*, and Jae-Woong Jeong*, “A temperature-responsive intravenous needle that irreversibly softens on insertion,” *Nature Biomedical Engineering* 8, 963–976, Oct. 2023. ([Link](#))
(Featured on the Cover)
9. Subin Oh, Sunwoo Lee, Sang-Hyuk Byun, **Simok Lee**, Choong Yeon Kim, Junwoo Yeo, Sein Chung, Shuo Li, Kyung-In Jang, Jiheong Kang*, and Jae-Woong Jeong*, “3D shape-morphing display enabled by electrothermally responsive, stiffness-tunable liquid metal platform with stretchable electroluminescent device,” *Advanced Functional Materials* 33, 2214766, Mar. 2023. ([Link](#))
(Featured on the Inside Back Cover)
10. **Simok Lee**†, Sang-Hyuk Byun†, Choong Yeon Kim, Sungwoo Cho, Steve Park, Joo Yong Sim, and Jae-Woong Jeong*, “Beyond human touch perception: An adaptive robotic skin based on gallium microgranules for pressure sensory augmentation,” *Advanced Materials* 34, 2204805, Nov. 2022. ([Link](#))
(Featured on the Front Cover)
11. Sang-Hyuk Byun†, Choong Sun Kim†, Karen-Christian Agno, **Simok Lee**, Zhuo Li, Byung Jin Cho*, and Jae-Woong Jeong*, “Design strategy for transformative electronic system toward rapid, bidirectional stiffness tuning using graphene and flexible thermoelectric device interfaces,” *Advanced Materials* 33, 2007239, Jan. 2021. ([Link](#))
(Featured on the Inside Back Cover)

Conference Presentations

1. **Simok Lee**, Min Seok Kim, Se-Yeon Heo, Yejin Ahn, Woo Jae Kim, Young Min Song, and Jae-Woong Jeong*, “Dual-Protected Physical Unclonable Functions Using Thermally Gated Liquid-Metal Composites”, *2026 Materials Research Society (MRS) Spring Meeting*, Apr. 2026
(Poster presentation)
2. Yejin Ahn, **Simok Lee**, and Jae-Woong Jeong*, “Bio-inspired Multi-Mode Pressure Sensor Enabled by Temperature-Programmable Liquid Metal-Polymer Composite”, *2026 Materials Research Society (MRS) Spring Meeting*, Apr. 2026
(Oral presentation)
3. **Simok Lee**, Yejin Ahn, Donghyun Lee, Sijoon Song, Yudeok Seo, Jae-Woong Jeong*, “Versatile Fine-Pitch Bonding with Low-Melting-Point Alloys for Advanced Electronic Integration”, *2025 International Symposium on Microelectronics and Packaging (ISMP)*, Nov. 2025
(Oral presentation)
4. Soomin Kim, Byungkook Oh, **Simok Lee**, Seok Hyun Hong, Jaewon Cho, Jae-Woong Jeong*, Steve Park*, “Ultrathin and highly Permeable Anti-Swelling and Non-Drying Hydrogel-Based Body-Temperature Phase-Transmittable Electrode for Stretchable Bioelectronics”, *The Polymer Society of Korea*, Sep. 2025
(Poster presentation)
5. Yejin Ahn, **Simok Lee**, Donghyun Lee, Sijoon Song, Yudeok Seo, Jae-Woong Jeong*, “Low-Temperature, Low-Pressure Anisotropic Conductive Resin with Low-Melting-Point Alloy for Next-Generation Displays”, *The 25th International Meeting on Information Display (IMID)*, Aug. 2025
(Poster presentation)

6. **Simok Lee**, Gun-Hee Lee, Seongjun Park, Steve Park, Jae-Woong Jeong*, “High-Resolution, Multifunctional Variable Stiffness Electronics by Solution-Processible, Chemically-Sintered Liquid Metal Ink”, *2025 Materials Research Society (MRS) Spring Meeting*, Apr. 2025
(Oral presentation)
7. Yejin Ahn, **Simok Lee**, Jae-Woong Jeong*, “Liquid Metal-Based Multi-Mode Pressure Sensor with Temperature-Dependent Sensitivity and Pressure Detection Range”, *2025 Korea Micro Electro Mechanical Systems Spring Conference*, Mar. 2025.
(Oral presentation)
8. Do A Kwon, **Simok Lee**, Jae-Woong Jeong*, “A Transformative Electronic Ink for Additive Manufacturing of Body-Temperature Softening Bioelectronics”, *2024 Materials Research Society (MRS) Fall Meeting*, Dec. 2024.
(Oral presentation)
9. **Simok Lee**, Gun-Hee Lee, Michael D. Dickey*, Jae-Woong Jeong*, “Conductivity Meets Transformability: Stiffness-tunable Electronics via Patternable Phase Change Metal Ink”, *2024 Carolina Science Symposium*, Nov. 2024.
(Poster presentation, All Scientific Poster Vendor Prize)
10. Mohammadreza Zare, Man Hou Vong, Mohamed Irfan, **Simok Lee**, Febby Krisnadi, Michael A. Daniele, Qingshan Wei*, Michael D. Dickey*, “Reversible Molecular Capture and Release with Liquid Metals”, *NCSU Schoenborn Research Symposium*, Oct. 2024
(Poster presentation)
11. **Simok Lee**, Gun-Hee Lee, Seongjun Park, Jae-Woong Jeong*, “Scalable High-resolution Printing of Mechanically-tunable Electronic ink for Transformative Bioelectronics”, *2024 12th World Biomaterials*, May 2024.
(Poster presentation)
12. Do A Kwon, **Simok Lee**, Jae-Woong Jeong*, “Body-temperature softening electronic ink for additive manufacturing of transformative bioelectronics via direct writing”, *2024 Spring Conference of The Korean Sensors Society & Sensor Expo Korea-Forum*, Apr. 2024.
(Poster presentation, Outstanding Paper Award)
13. Karen-Christian Agno, Keungmo Yang, Sang-Hyuk Byun, Subin Oh, **Simok Lee**, Heesoo Kim, Sungwoo Cho, Won-Il Jeong, and Jae-Woong Jeong*, “Softening Intravenous Needle through Body Temperature for Safer Patient Care and Injection Practice”, *International Conference on Nano/Micro Engineered & Molecular Systems*, May 2023.
(Oral presentation)
14. **Simok Lee**, Sang-Hyuk Byun, Jae-Woong Jeong*, “Uniform Gallium Microgranules-based Adaptive Electronic Skin for Advanced Pressure Sensing beyond Human Touch Perception”, *2023 Materials Research Society (MRS) Spring Meeting*, Apr. 2023.
(Poster presentation, Best Poster Award)
15. Karen-Christian Agno, Keungmo Yang, Sang-Hyuk Byun, Subin Oh, **Simok Lee**, Heesoo Kim, Kyurae Kim, Sungwoo Cho, Won-Il Jeong, and Jae-Woong Jeong*, “Thermally-Responsive Softening Intravenous Needle for Enhancement of Patient Care and Healthcare Provider Safety”, *2023 Materials Research Society (MRS) Spring Meeting*, Apr. 2023.
(Poster presentation)
16. **Simok Lee**, Sang-Hyuk Byun, Jae-Woong Jeong*, “Adaptive Electronic Skin with High

Sensitivity and Large Bandwidth based on Gallium Microdroplet-Elastomer Composite”, *Materials for Biomedical Systems Day*, Mar. 2023.

(Oral presentation)

17. Simok Lee, Sang-Hyuk Byun, Jae-Woong Jeong*, “Tunable skin surpassing pressure sensing ability of the human skin enabled by gallium microgranules-elastomer composite”, *International Conference on Active Materials and Soft Mechatronics*, Oct. 2022.

(Oral presentation)

18. Simok Lee, Sang-Hyuk Byun, Jae-Woong Jeong*, “Adaptive Pressure Sensor with High Sensitivity and Large Bandwidth Based on Gallium Microdroplet-elastomer Composite”, *2022 The Korean Sensors Society Autumn Conference*, Aug. 2022.

(Poster presentation, Best Paper Award)

19. Sang-Hyuk Byun, Choong Sun Kim, Karen-Christian Agno, Simok Lee, Zhuo Li, Byung Jin Cho, Jae-Woong Jeong*, “Gallium-based Transformative Electronic System Integrated with Graphene and Flexible Thermoelectric Device for Rapid Bi-directional Stiffness Tuning”, *2021 Materials Research Society (MRS) Spring Meeting*, Apr. 2021.

(Oral presentation)

Invited Talks

1. “Uniform Gallium Microgranules-based Adaptive Electronic Skin for Advanced Pressure Sensing beyond Human Touch Perception”, *31st International Materials Research Congress*, Aug. 2023.
(Fully-funded, selected as one of two presenters from 18 MRS Best Poster Award winners)
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Patents

1. Jae-Woong Jeong, Simok Lee, Sang-Hyuk Byun, Joo Yong Sim, “Adaptive Robotic Skin Based On Gallium Microgranules For Pressure Sensory Augmentation”, KR (PATENT-2023-0010818)
 2. Jae-Woong Jeong, Do A Kwon, Simok Lee, Steve Park, “Body-Temperature Softening Electronic Ink for Additive Manufacturing of Transformative Bioelectronics via Direct Writing”, KR (PATENT-2024-0111856)
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Other Activities

Teaching

1. KAIST Undergraduate Research Program (URP) Mentor, Jun. – Dec. 2025
- Project: "Omnidirectional Printing of Electronic Ink for Three-Dimensional Electronic Device"
2. Head Teaching Assistant, EE405 Electronics Design Lab, Spring 2025
3. Teaching Assistant, EE405 Electronics Design Lab, Fall 2021, Spring 2022, Fall 2023
4. Teaching Assistant, KAIST Youth Scientist Camp, Summer 2016, Summer 2017

Research Grants

1. Project leader for KAIST Venture Research Program, May – Dec. 2025 (\$8,500 USD)
- Project: "Development of reversible conductive adhesive for modular electronic device."

Service

1. Journal reviewer, Science Advances 2025
 2. Lead Organizer, 2021 Women Tech Stars Workshop (Supported by Google exploreCSR), Mar. – Aug. 2021
- Coordinated KAIST's first AI/CS/EE workshop for women in tech
 3. Mentor, KAIST Buddy-Mentor Program, Aug. – Dec. 2018
- Assisted international undergraduate exchange students
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References

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